Degree Project in Computer Science and Engineering

Second cycle, 30 credits

Adaptive Hybrid RAG

Optimizing the Cost-Performance Trade-off via Learned Query Routing

DANTE WESSLUND

$N = 150$

$N = 150$

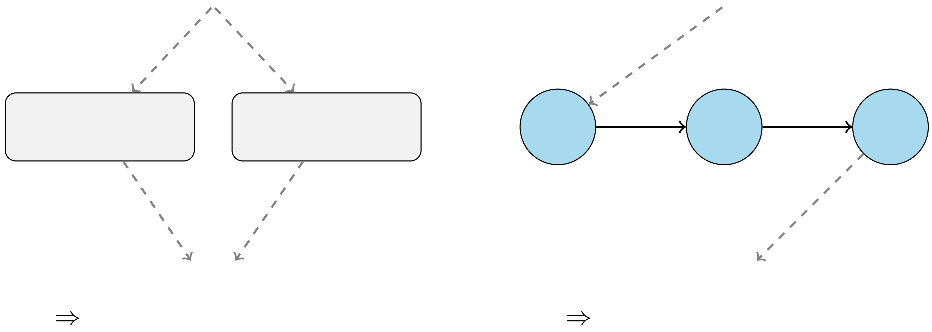
$q \mathcal{D}_k(q) k a$

$$p(a \mid q) = \sum_{d \in \mathcal{D}_k(q)} p_\theta(a \mid q, d)\, p_\eta(d \mid q),$$

$p_\eta(d \mid q) \eta p_\theta(a \mid q, d) \theta$

X

k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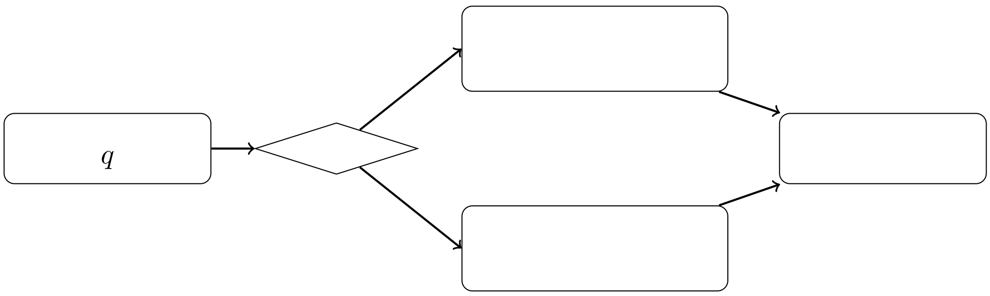
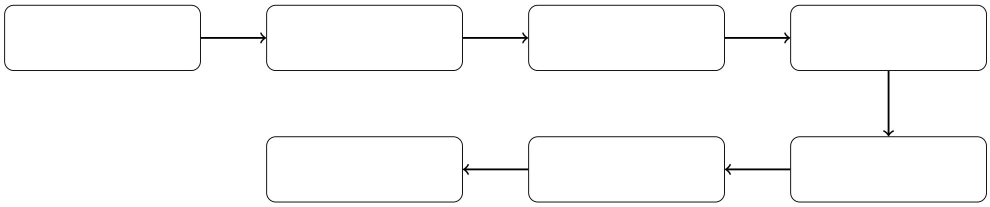

q



q_i

$$r_i^N = R_N(q_i), r_i^M = R_M(q_i),$$

$R_N R_M r_i^N r_i^M t_i^N t_i^M$
$N = 150 \bar{u}^N \bar{u}^M$

$r_i^N r_i^N r_i^M$
$y_i \in \{0, 1\}$

$$y_i = \begin{cases} 0, \\ 1, \end{cases}$$

2 000

X

$q_i s_\phi(q_i) \in [0,1] \phi P_\phi(y_i = 1 \mid q_i) y_i 1 q_i$

$V_i \in \mathbb{R}^V$

$$s_\phi(q_i) = \sigma({}^\top{}_i + b), \sigma(z) = \frac{1}{1 + (-z)},$$

$\in \mathbb{R}^V b \in \mathbb{R} L_2$

$h_i \in \mathbb{R}^d d$

$$z_i = W h_i + b, W \in \mathbb{R}^{2\times d}, b \in \mathbb{R}^2,$$

$z_{i,0} z_{i,1}$

$$s_\phi(q_i) = \frac{(z_{i,1})}{(z_{i,0}) + (z_{i,1})}.$$

$s_\phi(q_i) = \sigma(z_{i,1} - z_{i,0})$
n

$$\mathcal{L}(\phi) = -\frac{1}{n} \sum_{i=1}^{n} \alpha_{y_i} \ P_\phi(y_i \mid q_i),$$

$\alpha_c = n/(2\, n_c) c \in \{0,1\} n_c c$

$s_\phi(q_i)\tau \in [0,1]\pi_\tau$

$$\pi_\tau(q) = \begin{cases} M, s_\phi(q) \geq \tau, \\ N, s_\phi(q) < \tau, \end{cases}$$

$MNPR$

$$F_1 = \frac{2PR}{P+R}.$$

$r_i^N r_i^M t_i^N t_i^M \pi_\tau \tau$

$m\{\cdot\}10 r_M(\pi_\tau) = \frac{1}{m}\sum_{i=1}^m \{\pi_\tau(q_i) = M\}\pi_\tau C(\pi_\tau)$

$$C(\pi_\tau) = \frac{1}{m}\sum_{i=1}^{m} \left[\{\pi_\tau(q_i) = N\}\, t_i^N + \{\pi_\tau(q_i) = M\}\, t_i^M\right].$$

$U(\pi_\tau)\bar{u}^N \bar{u}^M$

$$U(\pi_\tau) = \left(1 - r_M(\pi_\tau)\right)\bar{u}^N + r_M(\pi_\tau)\,\bar{u}^M.$$

$$(\pi_\tau) = \frac{1}{m}\sum_{i=1}^{m} \{y_i = 1,\, \pi_\tau(q_i) = N\},$$

$$(\pi_\tau) = \frac{1}{m}\sum_{i=1}^{m} \{y_i = 0,\, \pi_\tau(q_i) = M\}.$$

$220{,}000 L_2 C = 1.01{,}000$

$2 \times 10^{-5} 0.0181610\,\% 1.0 \alpha_c 833$
71342

$\tau^\star [0.05, 0.95] \tau^\star$
$50\,\%$
$95\,\% 1\,000 \pm$

$\pi_\tau \tau$

$\tau 0 1 0.01$

$0.078 \pm 0.019 0.155 0.057 0.076 \pm 0.009$
0.6990.6670.7010.5850.8460.8030.7768,5317,13817,85752 %60 %

0.5000.25495 %1 000$\pm$

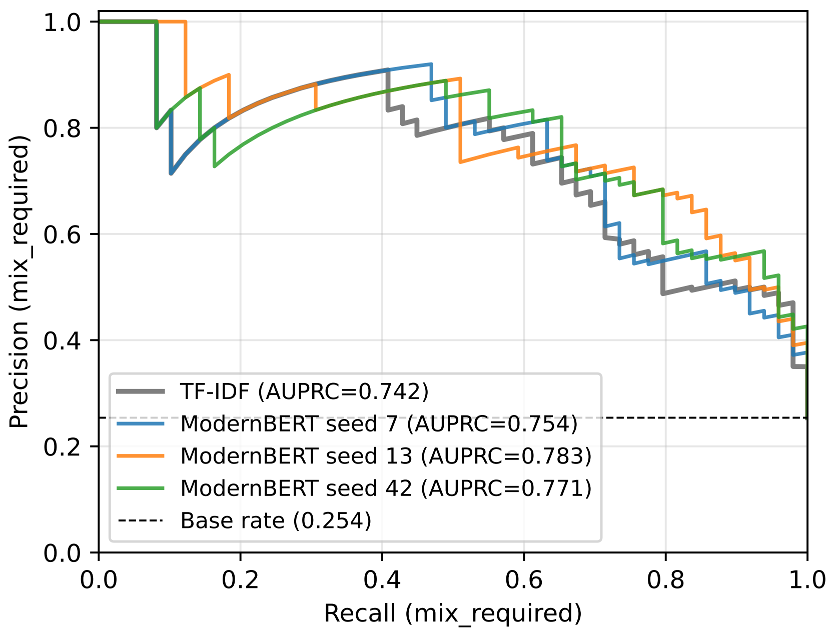
0.769 ± 0.015	0.908 ± 0.011	0.798 ± 0.012	0.701 ± 0.057	0.701 ± 0.042	0.699 ± 0.017	0.846 ± 0.016

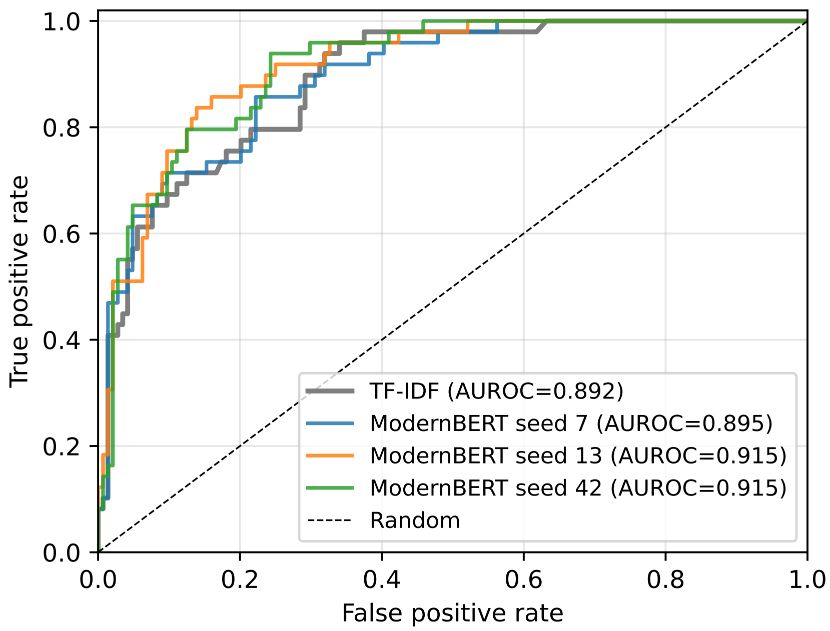
$$\bar{u}^N \bar{u}^M N = 150 (17{,}857 -)/17{,}857 \pm$$

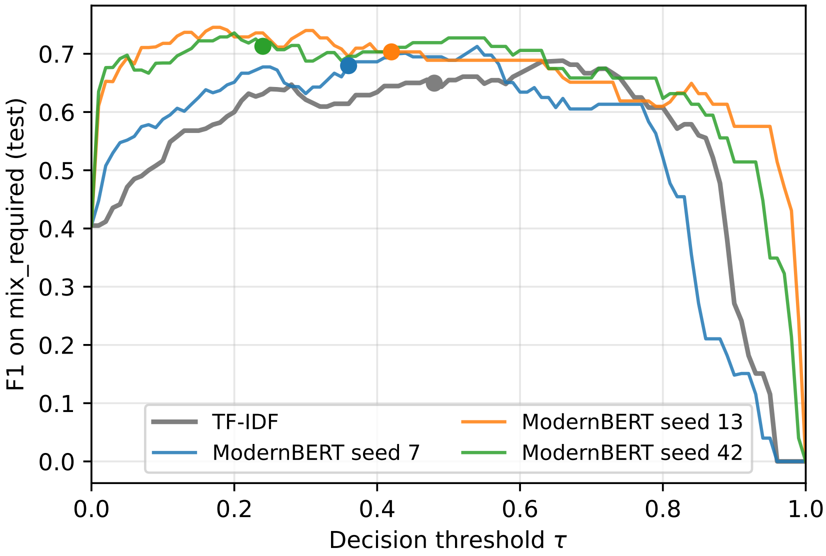
0.256 ± 0.027	14.92 ± 0.69		0.076 ± 0.009	0.078 ± 0.019

$\tau^\star = 0.24, 0.36, 0.42427130.6990.017$
$01 \tau = 0 \tau = 1$
τ
2908333865150.8390.8140.7990.7530.3750.1670.4180.3630.3160.2110.345 0.2730.7070.606

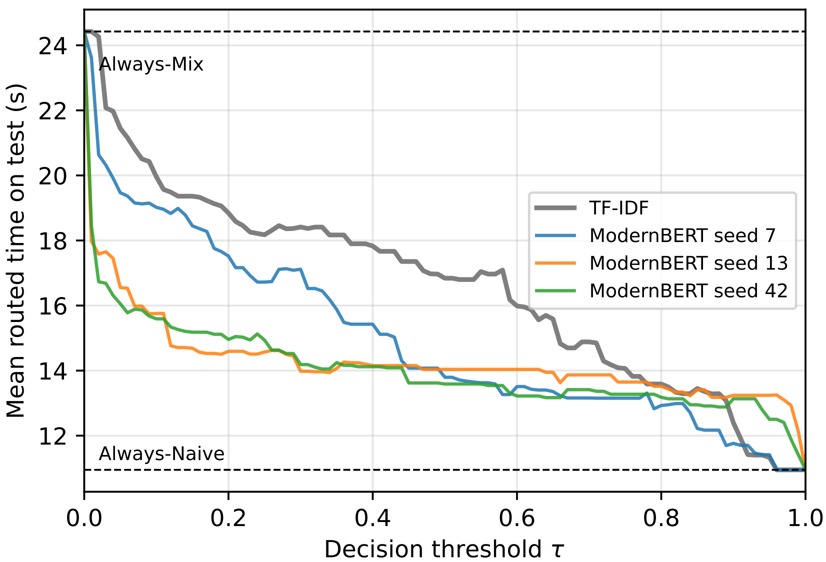
0.8390.8140.7530.7990.4180.316
0.667


TF-IDF (AUPRC=0.742)
ModernBERT seed 7 (AUPRC=0.754)
ModernBERT seed 13 (AUPRC=0.783)
ModernBERT seed 42 (AUPRC=0.771)
Base rate (0.254)
Precision (mix_required)
Recall (mix_required)
0.0
0.2
0.4
0.6
0.8
1.0

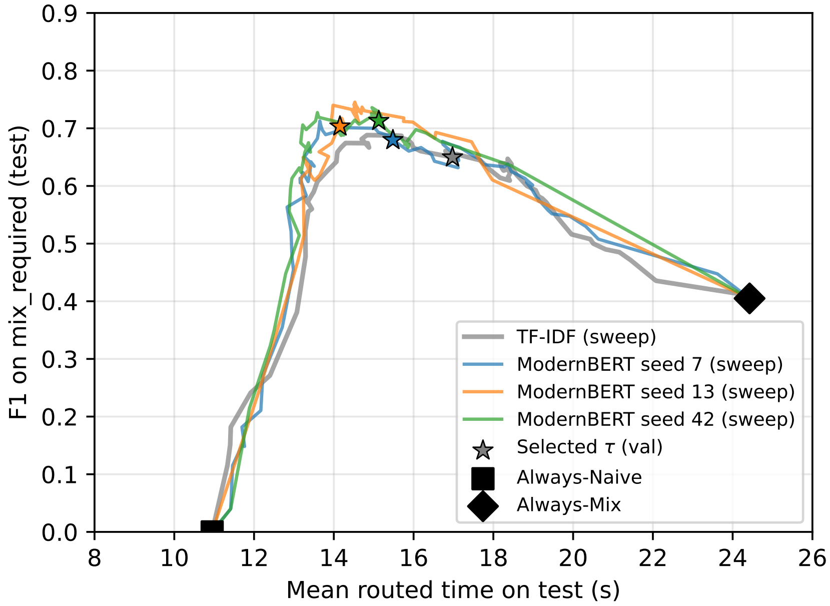

TF-IDF (AUROC=0.892)
ModernBERT seed 7 (AUROC=0.895)
ModernBERT seed 13 (AUROC=0.915)
ModernBERT seed 42 (AUROC=0.915)
Random
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0



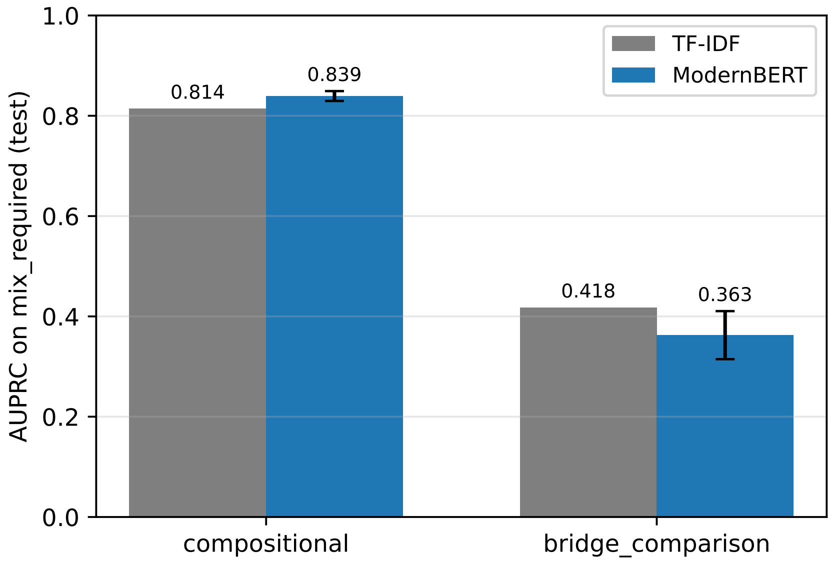
$$\tau^\star = 0.4842\tau^\star = 0.247\tau^\star = 0.3613\tau^\star = 0.42$$



$193[0.836, 0.936]0.908193 \leq 0.0170.0570.042193$
$10.9524.423{,}45817{,}85715173039\,\%7{,}1008{,}5005260\,\%$



F1 on mix_required (test)
Mean routed time on test (s)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
8
10
12
14
16
18
20
22
24
26
TF-IDF (sweep)
ModernBERT seed 7 (sweep)
ModernBERT seed 13 (sweep)
ModernBERT seed 42 (sweep)
Selected τ (val)
Always-Naive
Always-Mix



AUPRC on mix_required (test)
0.0
0.2
0.4
0.6
0.8
1.0
TF-IDF
ModernBERT
0.814
0.839
0.418
0.363
compositional
bridge_comparison

1930.7690.7420.7980.7840.6990.6500.8460.788$\leq$ 0.0170.0570.076 $\pm$ 0.0090.078 $\pm$ 0.0190.155

151710.9524.427,1008,5003,45817,85760 %52 %

0.6670.7940.803

$N = 150$1,286

1,286

193